Engine Model(s): PT6A

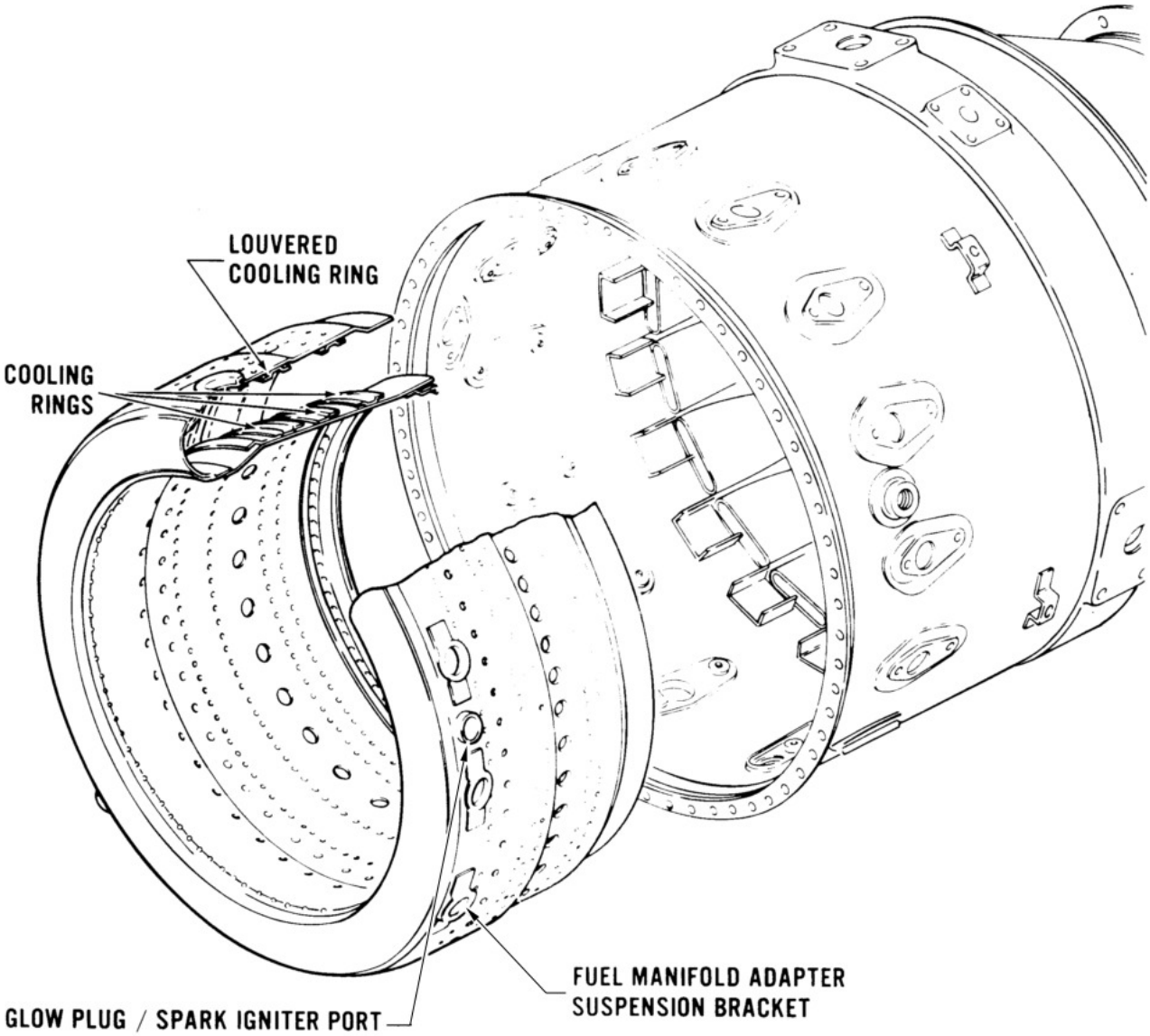
72-40-01

COMBUSTION CHAMBER LINER ASSEMBLY- DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Combustion Chamber Liner Assemblies



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The combustion chamber liner is an annular, heat-resistant steel liner, open at the rear end and domed at the front end. A series of straight, plunged and shielded holes in the inner and outer walls of the liner allow compressor discharge air to enter the liner combustion zone. Two glow plugs/ igniter sleeves and 14 fuel nozzle ports, which incorporate nozzle support brackets, are located in the liner outer wall adjacent to the domed end. The shape, size and locations of the holes and shields, as well as the locations of the glow plugs/igniter sleeves, provide the best fuel/air ratios for starting and sustained combustion and form efficient primary and secondary zones within the liner combustion zone. Primary air, introduced near the fuel nozzle ports, serves to support combustion. The secondary air provides a layer of cooling air adjacent to the liner inner walls; this effectively keeps the flame away from the walls and, by mixing, ensures that the gases leaving the liner are cooled to a temperature which the compressor turbine can tolerate; it also provides additional air for the combustion process.

The domed end of the liner is supported inside the front section of the gas generator case (Ref. 72-30-04) by the 14 fuel manifold adapter sheaths (Ref. 73-10-05) while the rear end of the liner is supported by the large and small exit ducts.

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